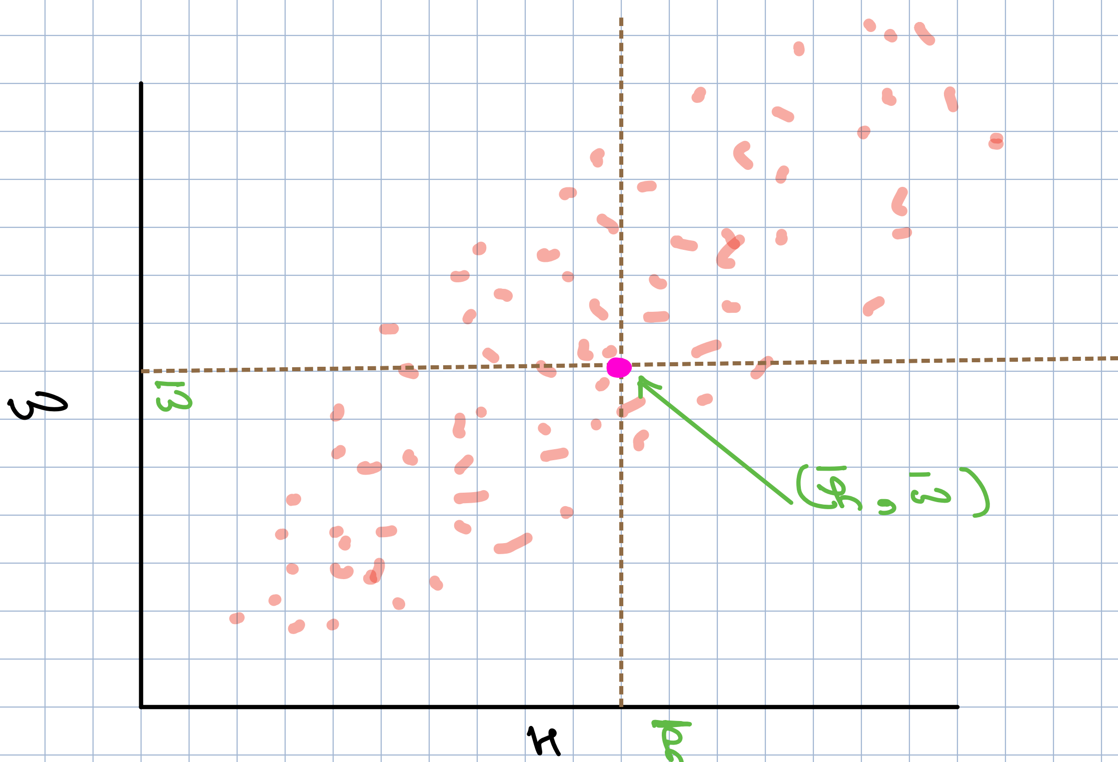
# Agenda

- Correlation
- Pearson (linear)
- Spearmann (Non-Linear)
- Visualizing correlation : Heatmap

Numerical vs Numerical

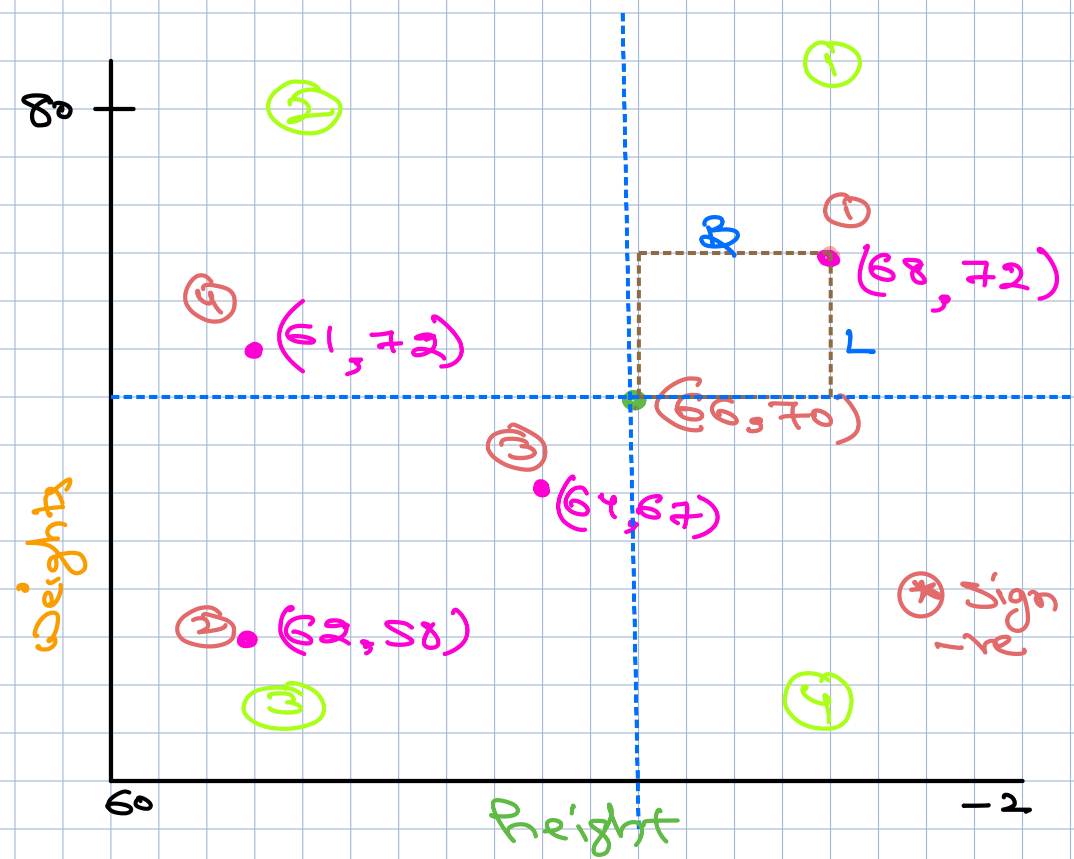
↓

Correlation



# Co-variance

| Height (inches) | Weight (kg) |
|---|---|
| 68 | 72 |
| 62 | 58 |
| 64 | 67 |
| 61 | 72 |
| 70 | 79 |
| 66 | 61 |
| 61 | 68 |
| 65 | 64 |
| 71 | 80 |
| 72 | 79 |
| $\bar{h}$ = 66 | $\bar{w}$ = 70 |




$(68-66) * (72-70) \Rightarrow 2 \times 2 \Rightarrow 4$ — 1st Q

$(62-66) * (58-70) \Rightarrow (-4) * (-12) \Rightarrow 48$ — 3rd Q

$(64-66) * (67-70) \Rightarrow (-2) * (-3) \Rightarrow 6$ — 3rd Q

$(61-66) * (72-70) \Rightarrow (-5) * 2 \Rightarrow -10$ — 2nd Q

$\Sigma$

⇒ Observation

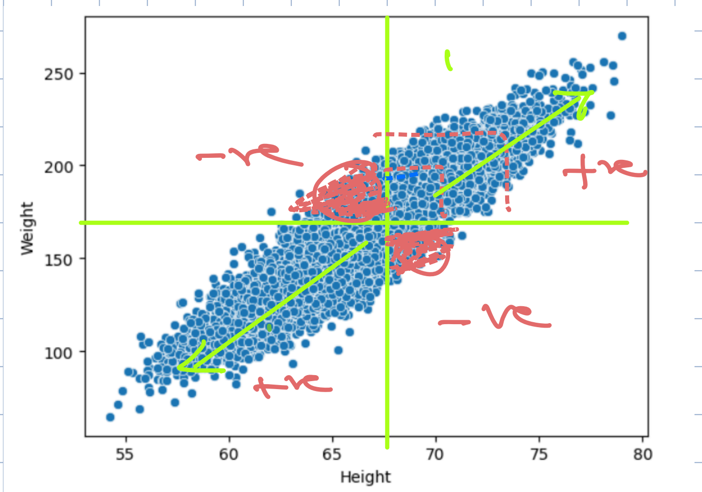
1. +ve area when points are in 1st and 3rd
2. −ve area when points are in 2nd and 4th

$$\boxed{Cov(x,y) \Rightarrow \frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x}) * (y_i-\bar{y})}$$

E or Cov(x, y)

↓

+ve



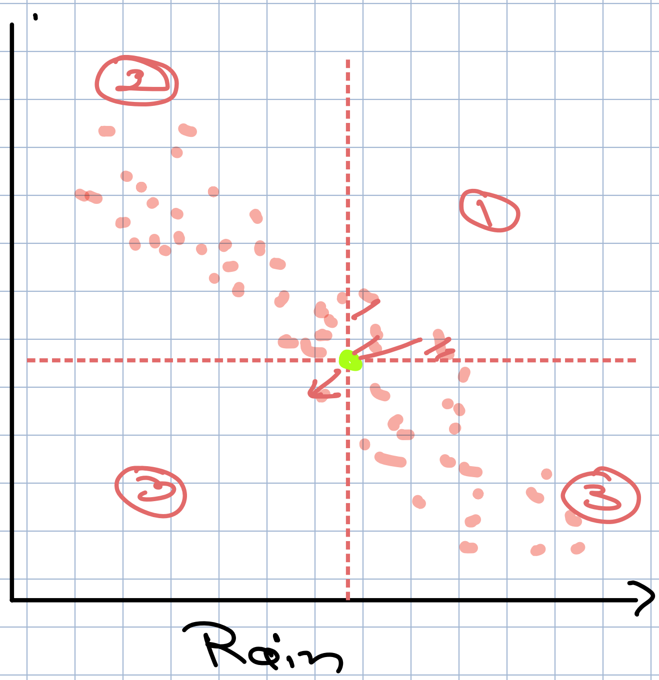

Ex 1

Cov(x, y)

↓

-ve

Ice Cream Sales

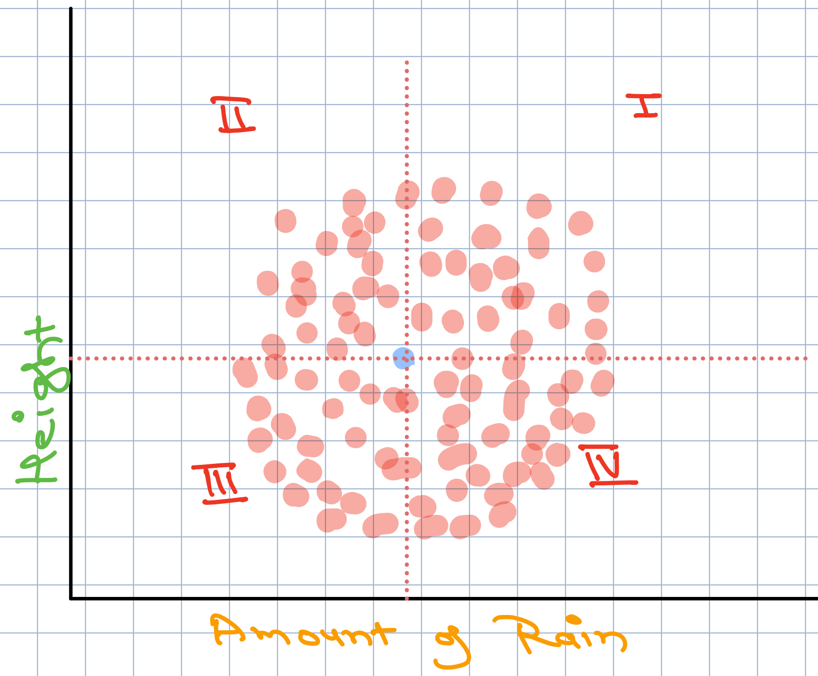



Ex -3

$Cov_{Male}(R, W) \longrightarrow +ve\ (9.22)$

$Cov_{Female}(R, W) \longrightarrow +ve\ (12.22)$

Ex - 3




Cov (R, P)

$\simeq 0$ ← small +ve −ve

* Covariance sign tells us about Direction of Relationship

* Magnitude tell us about strength

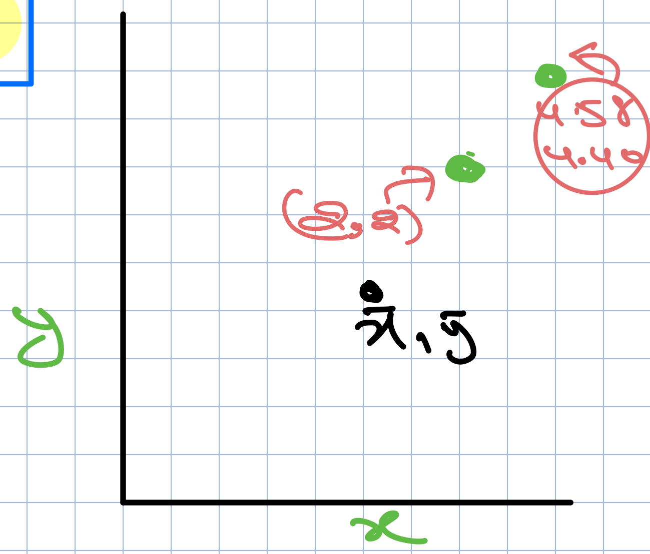
$Cov_{r_1} < Cov_{r_2}$

$r_1$ $r_2$

$\rho_1$ $\rho_2$

0.8 0.8

# Pearson Correlation




## Scenario

- Let's say researcher 1 measures H and W inches and Kg  $Cov_1$  $\sigma_1$
- while researcher 2 measures H and W cm and pounds  $Cov_2$  $\sigma_2$

1 inch = 2.54 cm
1 Kg = 2.20 p

## Questions

Does this Unit of measurement impact Variance?

- Sign will have no impact
- magnitude will change

Does the relationship actually changes across two Scenarios?

Relationship should be Same as both are working on same data

# Pearson Correlation

$$\boxed{\rho = \frac{\text{Covariance}(x, y)}{\sigma_x * \sigma_y}}$$

Range $(-1, 1)$

$$\frac{\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x}) * (y_i - \bar{y})}{\sigma_x * \sigma_y}$$

for H and ω :

i.e $\rho = \frac{1}{n}\sum_{i=1}^{n}(h_i - \bar{h})(\omega_i - \bar{\omega})$

## Intuition

① Cor is 1 → Strong positive Relation

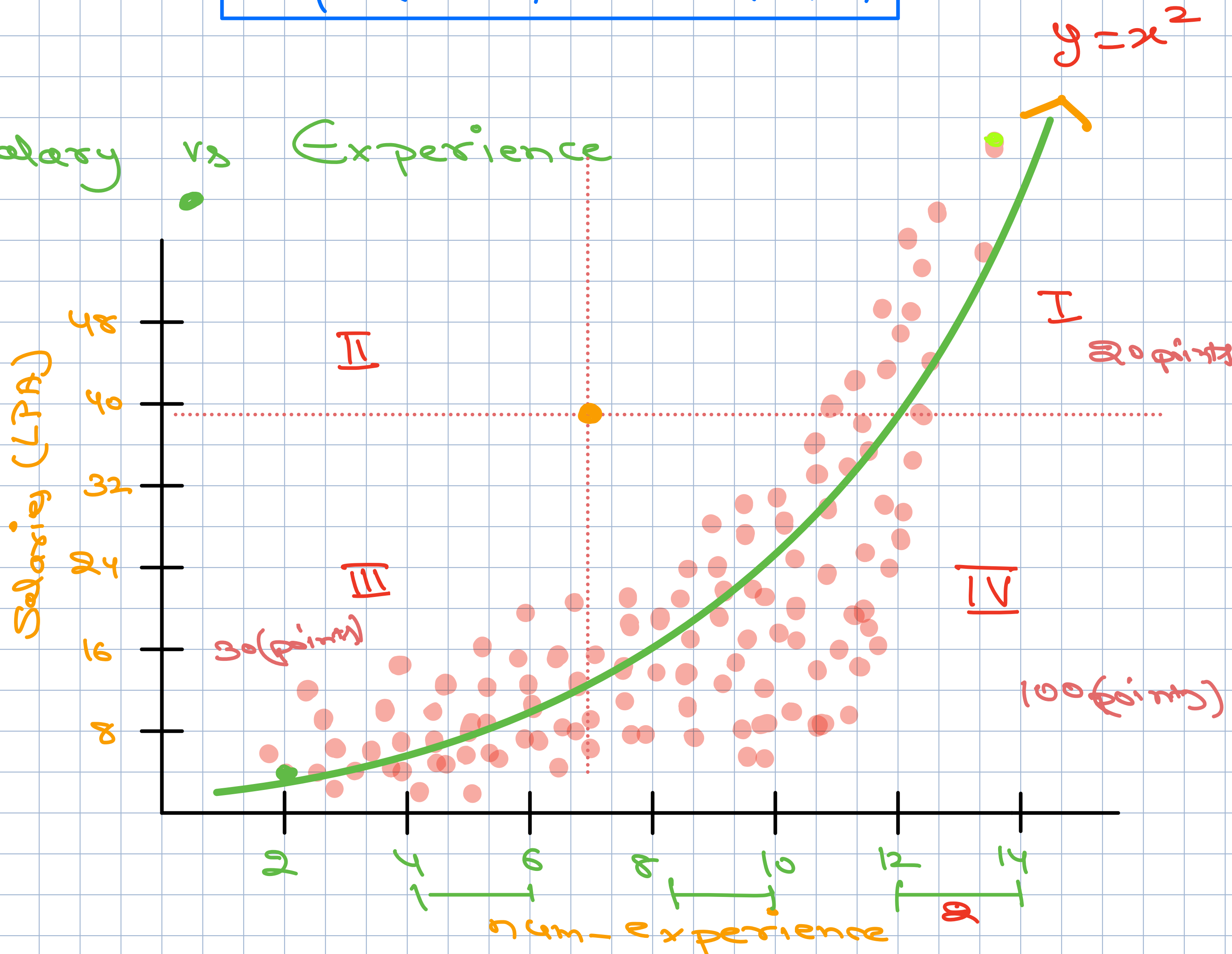
② Cor is -1 → Strong Negative Relation

③ Cor is 0 → No Relation

- By dividing covariance with std's we normalized the measurement (-1, 1) and made it Unit Less

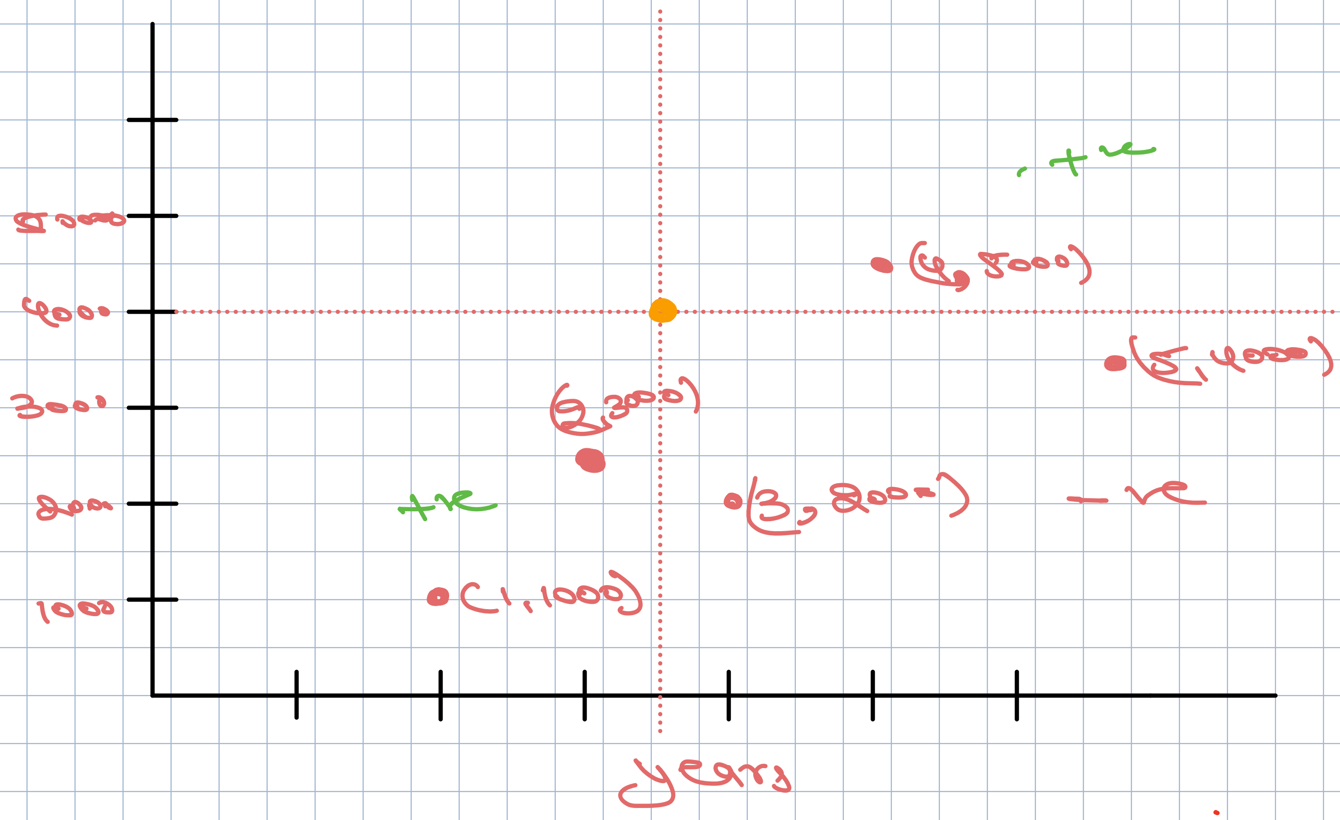
# Spearman Correlation

Salary vs Experience




Pearson correlation fails to Capture Non-Linear Relationship

Non-linear Relationships Spearmann correlation works better




* Spearman Correlation Tries to Solve the problem of Non-Linearity by using Ranks

| Experience | Rank_Exp | Salary | Rank_Sal | di |
|---|---|---|---|---|
| 1 | 1 | 1000 | 1 | 0 |
| 2 | 2 | 3000 | 3 | -1 |
| 4 | 4 | 5000 | 5 | -1 |
| 5 | 5 | 4000 | 4 | 1 |
| 3 | 3 | 2000 | 2 | 1 |

$$\rho_{\text{Spearman}} = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)}$$

Range $(-1, 1)$

$d_i$ = difference b/w two ranks of Obs $i$

$n$ = num- observation

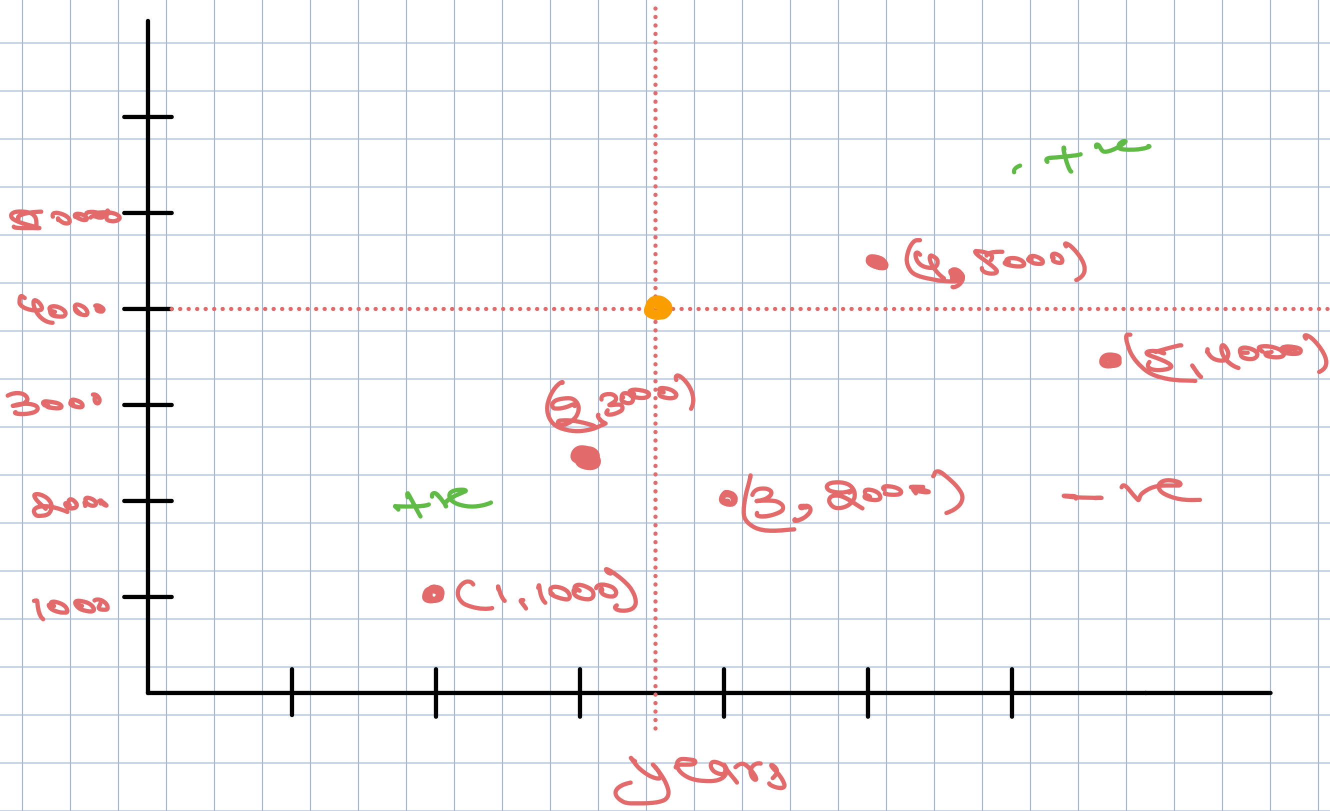
| X Experience (years) | Rank (Experience) | Y Salary ($) | Rank (Salary) | $d_i(RankX_i - RankY_i)$ | $d_i^2$ |
|---|---|---|---|---|---|
| 1 | 1 | 1000 | 1 | 0 | 0 |
| 3 | 3 | 2000 | 2 | 1 | 1 |
| 4 | 4 | 5000 | 5 | $-1$ | 1 |
| 5 | 5 | 4000 | 4 | 1 | 1 |
| 2 | 2 | 3000 | 3 | $-1$ | 1 |

$$\sum d_i^2 = 0+1+1+1+1 = 4$$

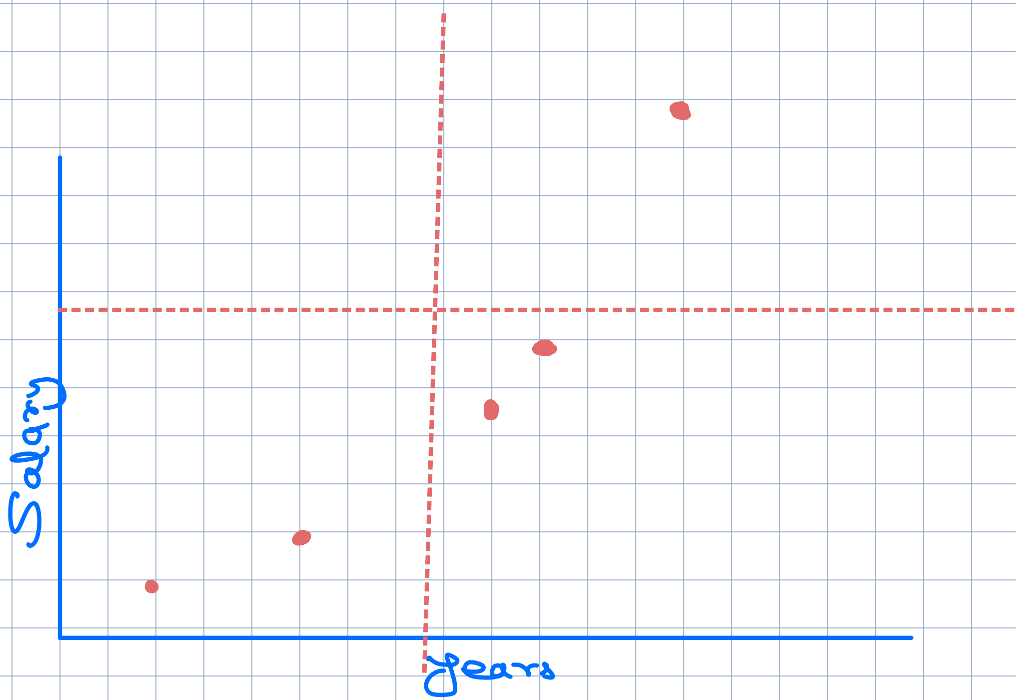
$$\rho = 1 - \frac{6\sum d_i^2}{n(n^2-1)} = 1 - \frac{6 \times 4}{5(5^2-1)} = 1 - \frac{1}{5} = 1 - 0.2 = 0.8 \rightarrow$$

Pearson $\approx 0$

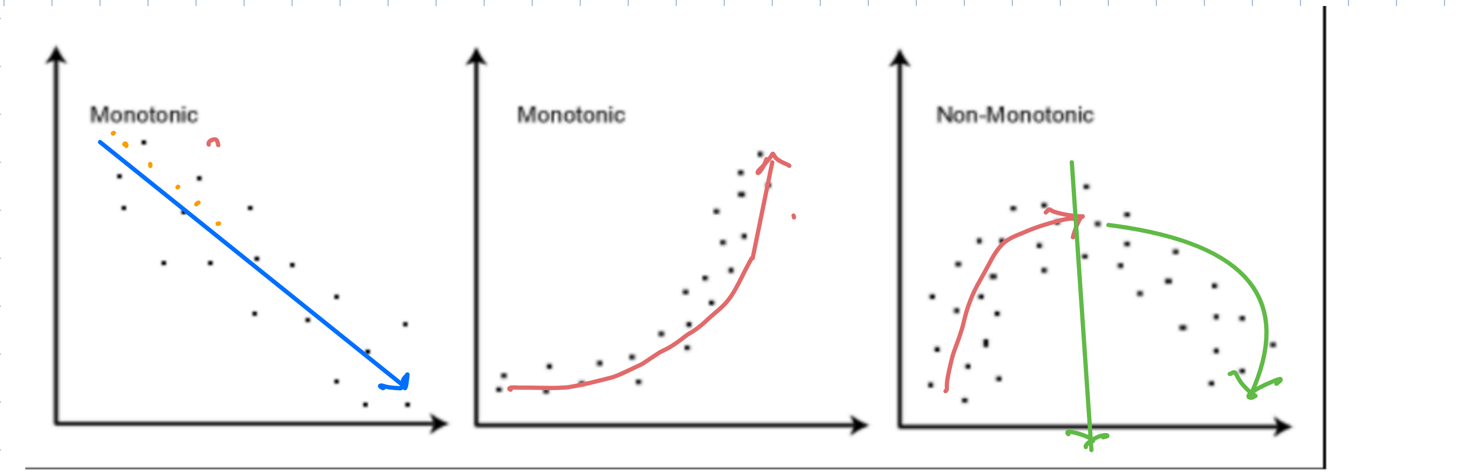
Spear $\approx 0.8$




Ranked Co-ordinates



# Monotonic vs Non-monotonic




Continously Decreasing

Continously Increasing

increase for First Half / Decrease in Second Half

* In monotonic relationship the trend is consistent

increasing

Decreasing

* Non-monontic → mix of trends

# Heatmap

① Color Density Represents Strength of Relationship

$H_0$ : No Relationship among vars

$H_a$ : Some Relationship among vars

P_corr , P_value

Correlation( x, y )

P_corr → Strength of Relationship

P_value → Significance of Relationship

Stage 0 0.001

P_val 0 0.08

P_val 0 0.45